

Selenium is a non-metallic photoconductive substance. It demonstrates photovoltaic action. Gray selenium is a semiconductor showing appreciable photoconductivity. It becomes more electrically conductive when it absorbs electromagnetic radiation or light. Selenium exists in several allotropes that interconvert upon heating and cooling carried out at different temperatures and rates. Element number 34, Selenium was discovered by Swedish chemist Jons Jacob Berzelius in 1817.

34. The most commonly used substance in fluorescent tubes are –

- (a) Sodium Oxide and Argon
- (b) Sodium Vapor and Neon
- (c) Mercury Vapor and Argon
- (d) Mercuric Oxide and Neon

U.P.P.C.S. (Pre) 2006

Ans. (c)

Fluorescent lamp tube is generally filled with a gas containing low pressure mercury vapor and Argon. Sometimes gases like Xenon, Neon or Krypton can also be used. The pressure inside the lamp is around 0.3% of atmospheric pressure.

35. Which gas is filled in normal tubelights?

- (a) Sodium vapour with argon
- (b) Sodium vapour with neon
- (c) Mercury vapour with argon
- (d) Mercury vapour with neon

M.P. P.C.S. (Pre) 1993

Ans. (c) & (d)

The normal tubelights are filled with mercury vapour with inert gas like argon or neon. An electric current in the gas excites mercury vapour which produces short wave ultraviolet light that causes phosphor coating inside the bulb to glow.

36. Which gas is filled at low pressure in a tubelight?

- (a) Argon and Neon
- (b) Neon and Mercury vapour
- (c) Nitrogen and Neon
- (d) Only Argon

Uttarakhand P.C.S. (Pre) 2005

Ans. (b)

See the explanation of above question.

37. Tubelight is filled with –

- (a) Sodium vapour
- (b) Argon gas at low pressure
- (c) Mercury vapour at low pressure
- (d) Mercuric oxide & Argon gas

U.P. Lower Sub. (Pre) 2009

Ans. (b) & (c)

See the explanation of above question.

38. The gas used in discharge tubes for optical decoration and advertising is –

- (a) Carbon dioxide
- (b) Ammonia
- (c) Sulphur dioxide
- (d) Neon

U.P.P.C.S. (Pre) 2001

Ans. (d)

The neon gas is used in discharge lamps, tubes and in fluorescent bulbs.

D. Acid, Base and Salt

Notes

Acid :

- The acid name comes from Latin acidus which means sour and refers to the sharp odour and sour taste of many acids.
- According to **Arrhenius**, acid is a compound that can release a proton or hydrogen ion H^+ in water.
- According to **Bronsted - Lowry** concept, acid is defined as any hydrogen containing material which can donate a proton to other substances.

Properties of Acids :

- (i) Acids in solution have a pH below 7.0.
- (ii) Taste is sour.
- (iii) Releases hydrogen ion (H^+) in water.
- (iv) Turns blue litmus paper into the red.
- (v) Chemical or substances having the property of an acid are said to be acidic.

Classification of acids :

- Acids are divided into two main classes :
- I. Strong acids :** These are very corrosive and can cause severe skin burns. Examples are hydrochloric acid (HCl), sulphuric acid (H_2SO_4) and nitric acid (HNO_3).
- II. Weak acids :** These are mildly corrosive and normally do not affect the skin. Examples are citric acid, acetic acid, tartaric acid etc.

Sources of Some Natural Acids

S.N.	Name of Acid	Source
1.	Tartaric Acid	Grapes, Tamarind, Citrus fruits, Bananas
2.	Acetic acid (Ethanoic acid)	Vinegar
3.	Citric acid	Lemon, Orange and commonly in all citrus fruits
4.	Oxalic acid	Tomato, Spinach, Broccoli
5.	Lactic acid	Curd
6.	Malic acid	Apple
7.	Ascorbic acid (Vitamin C)	Amla, Guava, Papaya, Orange
8.	Formic acid	Stings of ants and Stinging Nettle (a plant)

Uses of Acids :

- Tartaric acid, lactic acid, acetic acid and malic acid are edible acids.
- Oxalic acid is used to remove the rust on clothes.
- Conc-Nitric acid is used in the preparation of explosive substances and for purification of silver and gold.
- Aqua regia is a mixture of three-part of conc-hydrochloric acid and one part of conc-nitric acid (3 conc-HCl : 1 conc-HNO₃). It has capacity to dissolve platinum and gold. It is also known as **Royal water**. It is primarily used to produce **chloroauric acid** for refining the highest quality of gold. It is also used in laboratories to clean glassware of organic compounds and metal particles.
- Carbonic acid is one of the most common acid additives that is widely used in soft drinks such as Coca-cola.
- Certain acids are used as drugs. Acetylsalicylic acid (Aspirin) is used as a pain-killer and for bringing down fever.

Role of acids in human body :

- Acids play important roles in the human body.
- The hydrochloric acid present in the stomach helps in the digestion.
 - Amino acids are the structural unit of protein. For the synthesis of protein, amino acids are required. Protein is very essential for the growth and repair of body cells.

- Fatty acids are also required for growth and repair of body tissue.
- Nucleic acids (DNA and RNA) are essential for all known forms of life. DNA (deoxyribonucleic acid) and RNA (ribonucleic acid) are responsible for transmitting of traits to offsprings through genes and protein synthesis respectively.
- Carbonic acid is essential for the maintenance of pH equilibrium in the body.

Base :

- According to **Arrhenius**, a base is a compound that dissociates in water to form hydroxide OH⁻ ion.
- According to **Bronsted-Lowry** concept, base is any substance that can accept a proton from any other substance.

Properties of Bases :

- Bases have a bitter taste.
- Bases feel slippery.
- Bases are the conductor of electricity when dissolved in water, because they consist of charged particles in solution.
- A soluble base is called an alkali if it contains and releases OH⁻ ions quantitatively.
- Bases in solution have a pH above 7.0.
- Alkali turns red litmus into the blue.

Some bases are as follows :

- Calcium hydroxide Ca(OH)₂—Limewater
- Ammonium hydroxide NH₄OH—to clean glass windows.
- Sodium hydroxide NaOH—used to prepare soap.
- Potassium hydroxide KOH – used to prepare soap.
- Magnesium hydroxide Mg(OH)₂ – Its aqueous suspension is known as **milk of magnesia**, a common antacid. It is used to remove the acidity of the stomach and also as a laxative.

pH Scale :

- In chemistry, pH is a logarithmic scale used to specify the acidity or basicity (alkalinity) of an aqueous solution. It approximates the negative of the base 10 logarithm of the molar concentration, measured in units of moles per litre, of hydrogen ions in a solution.

- A pH value is a number from 1 to 14 with 7 as the middle (neutral) point. Values below 7 indicate acidity which increases as the number decreases while values above 7 indicate alkalinity which increases as the number increases.
- pH stands for potential of hydrogen.
- Pure water is neutral. Its pH value is 7.

Chemical Indicator :

- Any substance that gives a visible sign, usually by a colour change, of the presence or absence of any substance such as an acid or an alkali in a solution.
- Turmeric, litmus and shoe flowers (Gurhal) petals are natural chemical indicators.

Natural Indicator - Litmus (a natural dye) :

- Litmus is a water-soluble mixture of different dyes extracted from lichens. It is often absorbed onto filter paper to produce one of the oldest forms of pH indicator, used to test materials for acidity and basicity.
- It is available in the form of red and blue litmus paper.
- Acids turn blue litmus into red while alkali solutions turn red litmus into blue.
- Shoe flower turns the acidic solution into magenta while the basic solution is turned into green.

Acid-Base Indicators

S. No.	Name of Indicator	Change with acid	Change with base
1.	Blue litmus	Red	No change
2.	Red litmus	No change	Blue
3.	Turmeric	No change	Turns into red
4.	Methyl orange	Turns into red	Turns into yellow
5.	Phenolphthalein (Colourless)	No change	Turns into pink

Question Bank

1. Antacids are –

- (a) Base
- (b) Acid
- (c) Salt
- (d) Buffer
- (e) None of these

Chhattisgarh P.C.S. (Pre) 2016

Ans. (a)

Antacids are bases. They are formed by aluminum hydroxide, magnesium hydroxide, magnesium carbonate, magnesium trisilicate, calcium carbonate etc. It is used as a medicine to reduce excessive acid formed in stomach or prevent it from forming.

2. Goldsmiths uses aqua regia, which is prepared by mixing –

- (a) Nitric acid and Sulphuric acid
- (b) Nitric acid and Hydrochloric acid
- (c) Sulphuric acid and Hydrochloric acid
- (d) Citric acid and Benzoic acid

U.P.P.C.S. (Mains) 2014

Ans. (b)

Aqua regia is an acidic, corrosive and oxidative mixture of three parts concentrated hydrochloric acid (HCl) and one part concentrated nitric acid (HNO₃). It is called aqua regia because it is one of the few acid mixtures that can dissolve the 'noble' metals- Gold (Au), Platinum (Pt) and Palladium (Pd).

3. pH value shows the Numeric value of –

- (a) Quality of chemical used for developing photographic negatives
- (b) Analysing the acidic or basic nature of a solution
- (c) Analysing the intensity of an earthquake
- (d) Analysing for checking purity of milk

M.P.P.C.S. (Pre) 1996

Ans. (b)

The pH scale measures the acidity or basicity of a solution. The solution with a pH less than 7 is said to be acidic while the solution with the pH greater than 7 is basic or alkaline. The pure water has a pH value 7. The logarithm of the reciprocal of hydrogen ion concentration in moles per litre is measured by pH scale from 0 to 14.

4. The pH of a solution changes from 3 to 6. The H⁺ ion concentration will :

- (a) increase 3 times
- (b) decrease 3 times
- (c) decrease 10 times
- (d) decrease 1000 times
- (e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (d)

pH is a logarithmic scale used to specify the acidity or basicity (alkalinity) of an aqueous solution. It inversely indicates the concentration of hydrogen ions (H⁺) in the solution as it approximates the negative of the base 10 logarithm of the molar concentration of hydrogen ions (H⁺) in a solution. Each number represents a 10-fold change in the acidity/basicity of the solution. If the pH number is increased by 1, it means H⁺ ion concentration is decreased by 10 times. Hence, if the pH of a solution changes from 3 to 6, the H⁺ ion concentration will decrease 1000 times, and it will become less acidic.

5. Which of the following has pH value 7?
 (a) Pure Water (b) Neutral solution
 (c) Basic solution (d) Acidic solution
 (e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (e)

The pH value of pure water and neutral solution is 7. In general, water with pH lower than 7 is considered acidic and greater than 7 is considered basic. Pure water is considered neutral.

6. Which among the following solutions has pH value more than 7?

- (a) Buffer solution (b) Basic solution
 (c) Acidic solution (d) Neutral solution
 (e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (b)

See the explanation of above question.

7. A solution turns red litmus blue. The pH of solution is :

- (a) 7 (b) 1
 (c) 5 (d) 6.5
 (e) None of the above

Chhattisgarh P.C.S. (Pre) 2014

Ans. (e)

When a red litmus paper is immersed in a base or alkaline, the red litmus paper turns blue indicating the given solution as alkaline/base. When a blue litmus paper is immersed in an acid, the litmus paper turns red from blue indicating acid. The solution having pH value less than 7 is acidic and the solution having pH value more than 7 would be basic. According to the question, the red litmus paper turns blue which shows alkaline/basic nature of the solution. Thus, it has the pH value more than 7.

8. You have been provided with three test tubes. One of them contains distilled water and the other two contain an acidic solution and a basic solution, respectively. Which of them will turn red litmus to blue?

- (a) Distilled water
 (b) Acid
 (c) Base
 (d) More than one of the above
 (e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (c)

See the explanation of above question.

9. Why does dry HCl gas not change the colour of dry litmus paper?

- (a) HCl gas acts as a dehydrating agent.
 (b) Blue litmus becomes dry in the presence of dry HCl gas.
 (c) No H_3O^+ ions are present. Litmus changes colour only in the presence of H_3O^+ ions.
 (d) More than one of the above
 (e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (c)

All acidic aqueous solutions contain protonated water, known commonly as the hydronium ion (H_3O^+). Dry HCl gas does not change the colour of dry litmus paper, because in the absence of water no hydronium (H_3O^+) ions are formed. As HCl is not in aqueous form nor is the litmus paper wet and thus the lack of presence of hydrogen (H^+) or hydronium (H_3O^+) ions leads to the litmus paper not changing its colour. However, it is to be noted that blue litmus paper becomes red due to higher concentration of $\text{H}^+/\text{H}_3\text{O}^+$ ions (i.e. in the case of acids), while red litmus paper becomes blue due to higher concentration of OH^- ions (i.e. in the case of bases).

10. Which of the following may be a possible pH value of an acidic solution?

- (a) 6 (b) 7
 (c) 8 (d) 9

Chhattisgarh P.C.S. (Pre) 2018

Ans. (a)

If the pH of a solution is less than 7, the solution is called acidic, if the pH is about 7 the solution is neutral and if the pH is greater than 7, the solution is called basic. In an acidic solution, the concentration of hydrogen ions (H^+) is greater than the concentration of hydroxide ions (OH^-).

11. Which acid is considered as a basic chemical in Industry?

- (a) H_2CO_3 (b) HNO_3
 (c) H_2SO_4 (d) HCL

43rd B.P.S.C. (Pre) 1999

Ans. (c)

In the chemical industry, the sulphuric acid (H_2SO_4) is considered as basic chemical. Sulphuric acid is mainly used in petrochemical industries. It is used to refine metals, reduce corrosion of steel, refine oil and manufacture fertilizers. It is also used for cleaning purposes in the laboratory.

12. Which of the following is present in maximum amount in acid rain?

- (a) HCl (b) HNO_3

- (c) H_2SO_4 (d) H_2CO_3
 (e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (c)

The sulphur dioxide and nitrous oxide emitted from atmospheric pollution, industrial effusions and other activities occurring in the nature reach the atmosphere and react with oxygen and cloud water and fall on earth as sulphuric acid (H_2SO_4) and nitric acid (HNO_3) respectively. This is called Acid rain. The amount of sulphuric Acid (H_2SO_4) in the acid rain is about 70-75% and it is the primary acid rain component.

13. The component(s) of acid rain is/are :

- (a) HNO_3 (b) H_2SO_4
 (c) CO_2 (d) Both (a) and (b)
 (e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (d)

See the explanation of above question.

14. Which acid is present in Acid Rain?

- (a) Benzoic acid (b) Acetic acid
 (c) Nitric acid (d) Oxalic acid
 (d) None of the above

Chhattisgarh P.C.S. (Pre) 2017

Ans. (c)

See the explanation of above question.

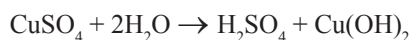
15. An aqueous solution of copper sulphate is acidic in nature because the salt undergoes –

- (a) dialysis (b) electrolysis
 (c) hydrolysis (d) photolysis

I.A.S. (Pre) 2001

Ans. (c)

Copper sulphate reacts with water which produces strong acid H_2SO_4 and weak base $\text{Cu}(\text{OH})_2$ as there is a process of hydrolysis. So its aqueous solution is acidic in nature.



16. Which is not a Lewis acid?

- (a) AlCl_3 (b) BF_3
 (c) NH_3 (d) FeCl_3

Jharkhand P.C.S. (Pre) 2013

Ans. (c)

The Lewis acids are lone pair acceptors. The Lewis bases are lone pair donors. Nitrogen in NH_3 has 5 electrons in which 3 of them are involved in bonding. That leaves one lone pair. This lone pair is available for bonding. Therefore, NH_3 is a Lewis base. AlCl_3 , BF_3 and FeCl_3 are Lewis acids.

17. Assertion (A) : To dilute sulphuric acid, acid is added to water and not water to acid.

Reason (R) : Specific heat of water is very high.

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
 (b) Both (A) and (R) are true, but (R) is not a correct explanation of (A).
 (c) (A) is true, but (R) is false.
 (d) (A) is false, but (R) is true.

I.A.S. (Pre) 1999

Ans. (a)

In order to dilute Sulphuric acid, acid is being added into water, which produces heat and energy. Water has a higher specific heat (heat capacity) than the acid, and so a vessel of cold water will absorb heat as acid is added. If water is added into acid then suddenly more heat would be generated and solution may boil very violently, splashing concentrated acid out of the vessel. So, acid is added to water to dilute sulphuric acid. Hence, both (A) and (R) are true and (R) is the correct explanation of (A).

18. Some metal oxides react with acid as well as base to produce salt and water. They are known as amphoteric oxides. The examples of amphoteric oxides are :

- (a) CuO and ZnO (b) Al_2O_3 and ZnO
 (c) Al_2O_3 and CuO (d) More than one of the above
 (e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (b)

Amphoteric oxides have both acidic and basic properties. The oxides of aluminum (Al_2O_3) and zinc (ZnO) are good examples. Other common examples of amphoteric oxides include BeO , MnO_2 , Cr_2O_3 , SnO and PbO . They react with acids and alkalis to form salts and water. The metals become cations in the former case and oxoanions in the latter. On the other hand, CuO is basic in nature, it is not amphoteric. Hence, option (b) is the correct answer.

19. Acids react with metal carbonates to liberate ____ gas.

- (a) H_2 (b) CO_2
 (c) CO (d) More than one of the above

(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (b)

Acids react with metal carbonates to liberate carbon dioxide (CO₂) gas. When an acid reacts with a metal carbonate, salt, water, and carbon dioxide are produced. Examples :



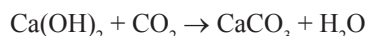
20. Limewater is turned milky by:

- (a) CO (b) CO₂
(c) O₂ (d) O₃
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (b)

Carbon dioxide (CO₂) reacts with calcium hydroxide to produce a white precipitate of calcium carbonate. Limewater is a solution of calcium hydroxide. So, if carbon dioxide is passed through limewater, the limewater will turn milky or cloudy white.



21. Sodawater obtained by passing carbon dioxide in water is :

- (a) An oxidising agent (b) Basic in nature
(c) Acidic in nature (d) A reducing agent

R.A.S./R.T.S.(Pre) 1999

Ans. (c)

The carbon dioxide (CO₂) gas dissolved in water can cause water to become acidic. The acidity of water from dissolved CO₂ can be reduced by a base such as baking soda (Sodium bicarbonate). When carbon dioxide is dissolved in the water, it forms carbonic acid (H₂CO₃) by reacting with water (H₂O) and remains in dissolved state at high pressure. So, the soda-water becomes acidic in nature.

22. Which of the following acid do not contain oxygen –

- (a) Nitric Acid (b) Sulphuric Acid
(c) Hydrochloric Acid (d) All of above

R.A.S./R.T.S. (Pre) 1992

Ans. (c)

Hydrochloric Acid (HCl) does not contain oxygen, as it is a compound of hydrogen and chlorine.

23. An unknown gas quickly dissolves in water. This gaseous aqueous solution turns red litmus into the blue. This gas produces white flame with hydrogen chloride.

This unknown gas is –

- (a) Sulphur dioxide (b) Nitric Hydroxide
(c) Ammonia (d) Carbon monoxide

R.A.S./R.T.S. (Pre) 1994

Ans. (c)

The aqueous solution of ammonia is basic due to forming of ammonium hydroxide which changes the colour of red litmus into blue. Ammonia produces a white flame when it reacts with hydrochloric acid.

24. The formula of baking soda is :

- (a) KHCO₃ (b) Na₂CO₃
(c) K₂CO₃ (d) NaHCO₃

U.P.P.C.S. (Pre) 2024

Ans. (d)

Sodium bicarbonate (sodium hydrogencarbonate), commonly known as baking soda, is a chemical compound with the formula NaHCO₃. Sodium bicarbonate is a white solid that is crystalline but often appears as a fine powder. In cooking, baking soda is primarily used in baking as a leavening agent. When it reacts with acid or is heated, carbon dioxide is released, which causes expansion of the batter and forms the characteristic texture and grain in cakes, quick breads, soda bread, and other baked and fried foods.

25. Match List-I with List-II and select the correct answer from the code given below the lists.

List-I	List-II
A. Bleaching powder	1. Sodium bicarbonate
B. Baking soda	2. Sodium carbonate
C. Washing powder	3. Calcium oxychloride
D. Plaster of Paris	4. Calcium sulphate hemihydrate

Code :

	A	B	C	D
(a)	3	4	2	1
(b)	2	3	4	1
(c)	3	1	2	4
(d)	4	3	1	2

U.P. R.O./A.R.O. (Mains) 2016

Ans. (c)

The correctly matched lists are as follows :

List-I	List-II
Bleaching powder	– Calcium hypochlorite/oxychloride $[\text{Ca}(\text{OCl})_2/\text{CaOCl}_2]$
Baking soda	– Sodium bicarbonate (NaHCO_3)
Washing powder	– Sodium carbonate (Na_2CO_3)
Plaster of Paris	– Calcium sulphate hemihydrate ($\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$)

26. Which one of the following pairs is not correctly matched?

- (a) Bleaching powder – Calcium oxychloride
 (b) Plaster of Paris – Calcium bicarbonate
 (c) Washing Powder – Sodium carbonate
 (d) Baking Soda – Sodium bicarbonate

U.P. R.O./A.R.O. (Mains) 2021

Ans. (b)

See the explanation of above question.

27. What is Neela thothea?

- (a) Copper Sulphate (b) Calcium Sulphate
 (c) Iron Sulphate (d) Sodium Sulphate

M.P.P.C.S. (Pre) 1990

Ans. (a)

Neela thothea (blue vitriol) is a compound of copper and sulphate. Its chemical formula is $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$.

28. The chemical used for destroying fungi in water tank is :

- (a) Nitric acid (b) Zinc sulphate
 (c) Magnesium sulphate (d) Copper Sulphate
 (e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (d)

Copper sulphate (blue vitriol or neela thothea) is used for destroying fungi in water tank. It is also used as a fungicide to protect field crops.

29. Blue vitriol is chemically :

- (a) Sodium sulfate (b) Nickel sulfate
 (c) Copper sulfate (d) Iron sulfate
 (e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (c)

See the explanation of above question.

30. Match the following lists by using codes given below.

Select the correct answer?

List-I	List - II
A. Washing soda	1. Sodium Hydroxide
B. Caustic soda	2. Copper Sulphate
C. Neela thothea	3. Sodium Thiosulphate
D. Hypo	4. Sodium Carbonate

Code :

	A	B	C	D
(a)	1	3	4	2
(b)	4	1	2	3
(c)	4	2	3	1
(d)	3	2	1	4

U.P. Lower Sub. (Pre) 2002

Ans. (b)

Washing soda	–	Sodium Carbonate
Caustic soda	–	Sodium Hydroxide
Hypo	–	Sodium Thiosulphate
Neela thothea	–	Copper Sulphate

31. Match the following lists by using codes given below.

Select the correct answer?

List-I	List - II
A. Neela thothea	1. Sodium bicarbonate
B. Epsom salt	2. Sodium Hydroxide
C. Baking soda	3. Magnesium Sulphate
D. Caustic soda	4. Copper Sulphate

Code :

	A	B	C	D
(a)	3	4	2	1
(b)	4	3	2	1
(c)	3	4	1	2
(d)	4	3	1	2

I.A.S.. (Pre) 1998

Ans. (d)

Neela thothea	–	Copper Sulphate
Epsom salt	–	Magnesium Sulphate
Baking soda	–	Sodium Bicarbonate
Caustic soda	–	Sodium Hydroxide

32. Baking soda is –

- (a) Sodium Chloride (b) Sodium Bicarbonate
 (c) Sodium Sulphate (d) Sodium Hydroxide

R.A.S./R.T.S. (Pre) 1993

Ans. (b)

Sodium bicarbonate is also known as Baking Soda. The chemical formula of Sodium bicarbonate is NaHCO_3 .

33. Baking Soda is –

- (a) Sodium Hydrogen Carbonate
- (b) Sodium Dihydrogen Carbonate
- (c) Disodium Hydrogen Carbonate
- (d) Sodium Carbonate
- (e) None of these

Chhattisgarh P.C.S. (Pre) 2016

Ans. (a)

See the explanation of above question.

34. The chemical formula of baking soda is :

- (a) $\text{Ca}(\text{OH})_2$
- (b) NaHCO_3
- (c) CaCO_3
- (d) Na_2CO_3

R.A.S./R.T.S. (Pre) 2016

Ans. (b)

See the explanation of above question.

35. Baking soda is –

- (a) Na_2CO_3
- (b) $\text{NaHCO}_3 \cdot 6\text{H}_2\text{O}$
- (c) NaHCO_3
- (d) $\text{NaCO}_3 \cdot 10\text{H}_2\text{O}$

Chhattisgarh P.C.S. (Pre) 2018

Ans. (c)

See the explanation of above question.

36. The chemical formula of washing soda is :

- (a) NaOH
- (b) Na_2CO_3
- (c) NaHCO_3
- (d) $\text{Ca}(\text{OH})_2$

43rd B.P.S.C. (Pre) 1994

Ans. (b)

Sodium carbonate is known as washing soda. It is used as detergent. Its formula is Na_2CO_3 .

37. What is the chemical name of Washing Soda?

- (a) Sodium carbonate
- (b) Sodium bisulphite
- (c) Sodium bi carbonate
- (d) None of the above

Chhattisgarh P.C.S. (Pre) 2023

Ans. (a)

See the explanation of above question.

38. Consider the following statements –

1. Anhydrous sodium carbonate is commonly known as baking soda.
2. Baking soda is used as a fire extinguisher.
3. Bleaching powder is produced in Hasenclever plant.

Which of these statements is right :

- (a) 1, 2 and 3
- (b) 2 and 3
- (c) 3 only
- (d) 1 and 2

I.A.S. (Pre) 2005

Ans. (b)

Sodium bicarbonate (NaHCO_3), also known as baking soda is used in bakery products, while anhydrous sodium carbonate (Na_2CO_3) is used as washing soda in soaps. Bleaching powder is prepared in Hasenclever plant or Backmann plant. Baking soda is also used as fire extinguisher equipment. Hence, statements 2 and 3 are correct.

39. Match List-I with List-II and select the correct answer using the codes given below the Lists :

List-I

List-II

- | | |
|-------------------------------|-----------------------|
| A. Artificial fruits ripening | 1. Carbon dioxide |
| B. Baking Soda | 2. Sodium silicate |
| C. Soft Drink | 3. Ethylene |
| D. Quartz | 4. Sodium bicarbonate |

Code :

- | | | | |
|-------|---|---|---|
| A | B | C | D |
| (a) 3 | 4 | 2 | 1 |
| (b) 3 | 4 | 1 | 2 |
| (c) 4 | 3 | 2 | 1 |
| (d) 2 | 1 | 4 | 3 |

U.P. P.C.S. (Mains) 2017

Ans. (b)

Ethylene is used in artificial fruit ripening, carbon dioxide is used in soft drinks, baking soda is the commercial name of sodium bicarbonate and quartz is used for producing sodium silicate.

40. Quartz crystals used in clocks are :

- (a) Sodium silicate
- (b) Silicon dioxide
- (c) Germanium oxide
- (d) Titanium dioxide
- (e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (b)

Quartz crystals used in clocks are composed of silicon and oxygen atoms. The atoms are linked in a continuous framework of SiO_4 (silicon-oxygen tetrahedra), with each oxygen being shared between two tetrahedra, giving an overall chemical formula of SiO_2 (silicon dioxide).

41. Which one of the following salt predominant in the ocean?

- (a) Calcium Carbonate
- (b) Sodium Chloride
- (c) Potassium Chloride
- (d) Magnesium Sulphate

Uttarakhand U.D.A./L.D.A. (Pre) 2003

Ans. (b)

Sodium chloride (NaCl) is known as common salt or table salt. It is found in ocean water (Salt water) and salt lakes. It is predominant salt in the ocean water. It is prepared by evaporating salt water.

42. Table salt (NaCl) is a product of –

- (a) Weak acid and weak base
- (b) Strong acid and strong base
- (c) Weak acid and strong base
- (d) Strong acid and weak basic

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (b)

The compounds formed after reaction between the strong base and strong acid are known as neutral salt. Sodium chloride (NaCl) is an example of neutral salt which is also known as Table Salt.

43. Which one of the following salts should not be used as dining table salt?

- (a) Iodized salt
- (b) Sea salt
- (c) Rock salt
- (d) Processed common salt

U.P.R.O./A.R.O. (Pre) 2014

Ans. (*)

According to the question, all the four above-mentioned salts can be used as table salt. Sea salt, rock salt and table salt contain around 100% sodium chloride which means that too much of any of them will have a negative effect on your health. However, rock salt and sea salt are often marketed as being a healthier or tastier option than table salt and can be considerably more expensive.

44. Iodised salt contains :

- (a) Free iodine
- (b) Calcium iodide
- (c) Magnesium iodide
- (d) Potassium iodide
- (e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (d)

Potassium iodide(KI) is similar to food salt in the structure and physical properties. This is commonly used in making iodised salt. In addition to potassium iodide, sodium iodide and sodium iodate are also used in making iodised salt.

45. Which one of the following statements is not correct ?

- (a) The presence of NaCl increases the rate of setting of Plaster of Paris
- (b) Gypsum is added to the cement to slow down its rate of setting
- (c) All alkaline earth metals form hydrated salts
- (d) Barium and Strontium are found free in nature

I.A.S. (Pre) 2003

Ans. (d)

Barium (Ba) and Strontium (Sr) are not found in nature as free element. These two belong to Group 2 of the periodic table. The metals of this particular group are highly reactive. As a result of which, when they come into contact with the atmosphere, they form compounds. Other three statements are correct.

46. What is untrue for bleaching powder?

- (a) It is more soluble in water
- (b) It is powder of light yellow colour
- (c) It is an oxidant
- (d) It releases chlorine after reaction with dilute acid

39th B.P.S.C. (Pre) 1994

Ans. (a)

Calcium Hypochlorite/Calcium Oxychloride is an inorganic compound which is the main ingredient compound of bleaching powder. Its chemical formula is $\text{Ca}(\text{OCl})_2$ / CaOCl_2 . It is a white solid, although commercial samples appear yellow. It is an oxidizing agent and releases chlorine after reaction with dilute acid. Bleaching powder is used for water treatment and acts as a bleaching agent. It is not highly soluble in water.

47. Which chemical compound is present in Bleaching Powder?

- (a) Calcium Hydroxide
- (b) Calcium Oxychloride
- (c) Calcium Carbonate
- (d) Ammonium Chloride

U.P.U.D.A./L.D.A. (Spl.) (Mains) 2010

Ans. (b)

See the explanation of above question.

48. When eno salt is poured into the water, bubbles forms, which is due to –

- (a) O₂ gas (b) CO₂ gas
(c) CO gas (d) H₂ gas

R.A.S./R.T.S. (Pre) 1999

Ans. (b)

When eno salt is dissolved in water, the carbon dioxide gas is produced which produces gas bubbles.

49. To develop photographic plates –

- (a) Sodium thiosulphate is used as oxidising agent.
(b) Free bromine releases.
(c) Light affected area quickly oxidizes.
(d) The solution is kept developing until all the bromide is not dissolved.

R.A.S./R.T.S. (Pre) 1994

Ans. (a)

Sodium thiosulphate is used in photographic processing as a fixer in photography. Sodium thiosulphate removes silver bromide as a soluble compound from the negative plate or film which is obtained at the end of photographic development. It is used as oxidising agent.

50. Which one of the following substances is used in photography and also as an antichlor ?

- (a) Chrome red (b) Sodium thiosulphate
(c) Hydrogen peroxide (d) Calomel

U.P.P.C.S. (Mains) 2010

Ans. (b)

Sodium thiosulphate (Hypo) is a colourless crystalline substance. It is used in photography and also as an antichlor. An antichlor is a substance used to decompose residual hypochlorite or chlorine after chlorine-based bleaching.

51. The chemical used as a 'fixer' in photography is –

- (a) Sodium sulphate
(b) Sodium thiosulphate
(c) Ammonium persulphate
(d) Borax

I.A.S. (Pre) 1995

Ans. (b)

Sodium thiosulphate (Hypo) is used in photography as a fixer of positive and negative. It is also used in the extraction of Gold and Silver.

52. The chemical used as fixer in the photography is –

- (a) Sodium thiosulphate (b) Borax

- (c) Sodium tetrathionate (d) Ammonium molybdate

U.P.R.O./A.R.O. (Pre) 2014

Ans. (a)

See the explanation of above question.

53. The hypo solution used in photography is an aqueous solution of –

- (a) Sodium thiosulphate (b) Sodium tetrathionate
(c) Sodium sulphate (d) Ammonium per sulphate

U.P.P.C.S. (Pre) 2010

Ans. (a)

See the explanation of above question.

54. The useful element in photography is –

- (a) Silver Nitrate (b) Silver Bromide
(c) Sulphuric Acid (d) Citric Acid

U.P.P.C.S. (Pre) 1992

Ans. (b)

To develop the photographic plate, the plate is kept in the solution of silver bromide until all the silver bromide is dissolved.

55. Which one of the following is coated on the photographic plate?

- (a) Silver oxide (b) Silver bromide
(c) Silver chloride (d) Silver iodide

U.P.P.C.S. (Mains) 2015

Ans. (b)

High resolution photographic plates have a coating of exceedingly fine-grained photosensitive emulsions with sub-microscopic crystals of silver bromide in gelatin.

56. Which one of the following substances is used in glazing the pottery?

- (a) Alum (b) Calomel
(c) Zinc chloride (d) Zinc oxide

U.P.P.C.S. (Mains) 2015

Ans. (d)

Zinc oxide is used in glazing the pottery. Here are some more oxides which are used in glazing- silicon dioxide, aluminium oxide, barium oxide, sodium oxide etc.

57. Which of the following imparts deep blue colour to glass?

- (a) Cobalt oxide (b) Cupric oxide
(c) Ferric oxide (d) Nickel oxide

U.P. R.O./A.R.O. (Mains) 2017

Ans. (a)